# Greenhouse Gas Report 2024

Annual atmospheric summary and regional carbon balance, 2010-2024

**Summary.** Annual concentrations of carbon dioxide ($CO_2$), methane ($CH_4$) and nitrous oxide ($N_2O$) increase in the constructed 2024 record. The regional carbon balance also reaches its highest total in the illustrated period. Concentration indicators describe atmospheric composition; regional fluxes describe annual carbon exchange. These measures use different units and accounting boundaries. This edition adds monthly patterns, reporting-zone allocations and sensitivity checks. All measurements, locations, coverage counts and interpretations are fictional.

## 1 Atmospheric concentrations

The annual mean $CO_2$ concentration increases from 424.4 to 427.6 parts per million (ppm). Methane rises from 1,968 to 1,978 parts per billion (ppb), while nitrous oxide rises from 340.1 to 341.2 ppb. Absolute changes retain the unit of each gas and should not be compared as though the three scales were interchangeable.

Relative changes use the 2023 mean as their denominator. $CO_2$ has the largest proportional increase in Table 1. These percentages are concentration changes, not shares of emissions or contributions to warming. Monthly detail on page 3 provides a separate view of seasonal variation without changing the annual reporting basis.

## 2 Coverage and comparability

The concentration dataset uses 24 fictional sampling locations and up to 12 monthly records per location. $CO_2$ has complete coverage; one full site-year is absent for methane and two are absent for nitrous oxide. Each gas uses the same retained locations in 2023 and 2024, avoiding a change in its comparison group.

Each retained location contributes an annual mean based on 12 monthly values. Location means receive equal weight. Coverage counts identify available site-month records, not independent samples or measurement precision.[2] Assigned uncertainty terms are reported separately; they do not establish whether a year-on-year difference is statistically significant.

**Table 1.** Annual mean concentrations, year-on-year changes and reporting coverage.

| Greenhouse gas | Unit | Annual mean concentration[1] | | Change, 2023-2024 | | Valid months[2] |
|---|---|---|---|---|---|---|
| | | 2023 | 2024 | Absolute | Relative | 2024 |
| Carbon dioxide ($CO_2$) | ppm | 424.4 ± 0.2 | 427.6 ± 0.2 | +3.2 | +0.75% | 288 / 288 |
| Methane ($CH_4$) | ppb | 1,968 ± 2 | 1,978 ± 2 | +10 | +0.51% | 276 / 288 |
| Nitrous oxide ($N_2O$) | ppb | 340.1 ± 0.1 | 341.2 ± 0.1 | +1.1 | +0.32% | 264 / 288 |

ppm = parts per million; ppb = parts per billion. Absolute changes use the unit in column 2. Relative changes are rounded to two decimal places.

## 3 Regional accounting boundary

The regional series covers one illustrative land area, held constant from 2010 through 2024. It separates fire releases from non-fire net land exchange, labelled net biome exchange (NBE). A negative NBE value represents uptake; a positive value represents release. The same sign convention applies to every year in Figure 1.

Regional totals exclude fossil-fuel and ocean fluxes. They are not a global emissions inventory. Concentrations in Table 1 are not used to calculate the regional balance, and the concentration and flux datasets remain independent.

$$\text{Total } CO_2 = \text{fire } CO_2 + \text{NBE}$$

The total already includes both components. Adding all three plotted series would count the regional balance twice.

## 4 Reporting conventions

The regional definitions below also apply to the zone allocations on page 4 and the scenario comparisons on page 5. They do not redefine the concentration units in Table 1.

**Table 2.** Regional series and sign conventions.

| Series | Meaning in this report |
|---|---|
| Total $CO_2$ | Net balance, including fire |
| Fire $CO_2$ | Gross carbon released by fire |
| NBE | Non-fire net land exchange |
| Negative value | Uptake exceeds releases |

Figures show carbon mass in GtC $yr^{-1}$, not $CO_2$ mass. Monthly shares are percentages of an individual zone's annual fire flux; they are not percentages of the combined regional total.

## Reading the annual record

The annual concentration and regional balance views answer different questions. A higher concentration establishes a higher value within a gas-specific series. A higher regional total establishes a larger net release within the fixed land boundary. Neither statement determines how another gas, zone or season must change. Compare like units and reporting periods before combining results.

## Boundary consistency

The additional detail in this edition is reconciled to the annual reference quantities. Seasonal concentration profiles preserve the reported annual means. Zone balances preserve the regional total, and monthly fire shares preserve each zone's annual fire component. The sensitivity cases are separate alternatives to those reference values, not additions to the inventory.

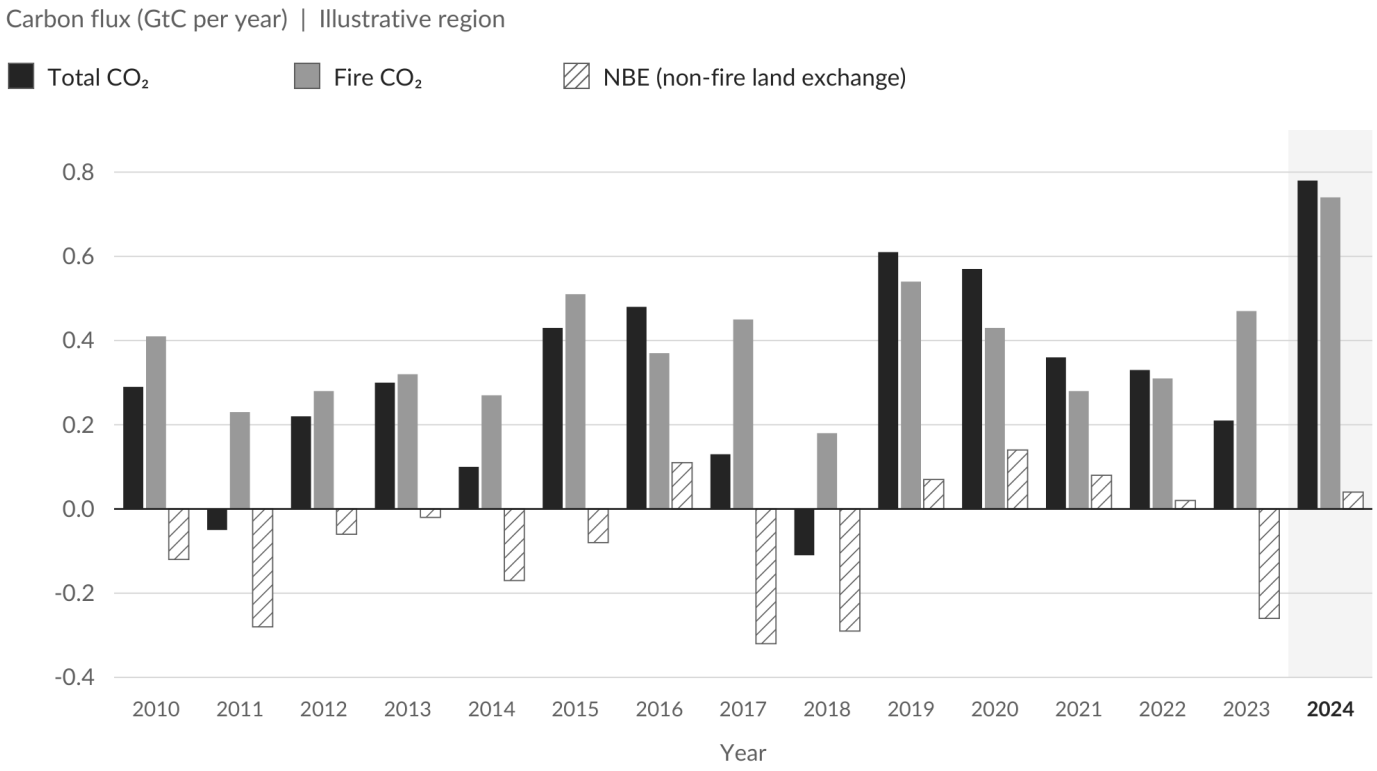
**Notes.** [1] Values after ± are assigned terms for the synthetic annual means, not empirically estimated confidence intervals. [2] Valid months count 24, 23 and 22 retained sites, respectively, multiplied by 12 months. No missing site-month values are imputed. Figure 1: page 2; seasonal detail: page 3; zone allocations: page 4; methods and sensitivity: page 5.

# Total $CO_2$ emissions and regional variability

**5 Annual regional balance.** Figure 1 compares total $CO_2$ emissions, fire emissions and NBE over 15 reporting years. The largest total occurs in 2024. Earlier years alternate between stronger uptake and additional non-fire release, so changes in fire do not translate into equal changes in the total.

The balance becomes negative when land uptake exceeds fire releases, as in 2011 and 2018. In 2024 both components are positive. Exact final-year values appear in the figure's summary labels. Read those values with the chart unit, not the concentration units on page 1.

## Annual regional $CO_2$ emissions, 2010–2024




| 2024 TOTAL $CO_2$ | 2024 FIRE $CO_2$ | 2024 NBE |
|---|---|---|
| **0.78 GtC** | **0.74 GtC** | **+0.04 GtC** |

**Figure 1.** Annual regional $CO_2$ balance, 2010-2024. Dark bars show total emissions, grey bars show fire releases and hatched bars show NBE. The shaded final group identifies 2024. Units are gigatonnes of carbon per year (GtC $yr^{-1}$); positive values are releases and negative values are uptake. All values are synthetic.

### 6 Interpreting the components

All bars share a common zero line. Fire remains positive in 2011 and 2018 even though the total is negative. Those years reflect an offset within the regional balance, not an absence of fire activity. Negative NBE also reduces the total in 2023, but does not reverse its sign.

Add fire emissions and NBE to recover the total. In the final year, a positive NBE contribution raises the total above the fire component. The largest fire year and the largest total year coincide here, but the accounting relationship does not require that pattern. Zone-level comparisons on page 4 retain this boundary and sign convention. The scenario exercise on page 5 changes selected component assumptions; it does not revise the original annual record.

### 7 Selected-year checks

**Table 3.** Component reconciliation (GtC $yr^{-1}$).

| Year | Fire | NBE | Total |
|---|---|---|---|
| 2011 | 0.23 | -0.28 | -0.05 |
| 2018 | 0.18 | -0.29 | -0.11 |
| 2023 | 0.47 | -0.26 | 0.21 |

Annual values were constructed directly and rounded to two decimal places. Totals were calculated from the components before plotting. The monthly fire allocations introduced on page 4 are a separate constructed disaggregation; they are not an observed seasonal record.

No uncertainty range is assigned to this regional series. Component arithmetic is exact at the published precision. Retain the negative signs when copying an uptake value; omitting a sign changes both the component meaning and the reconciled total.

**Scope.** NBE means non-fire net land exchange in this report. GtC is carbon mass, not $CO_2$ mass. The regional balance cannot be added to ppm or ppb concentrations or used to infer a global concentration change. No real location, measured trend or causal attribution is implied.

# Seasonal concentration patterns and coverage

Annual means conceal variation within a reporting year. The monthly concentration series below is a constructed extension of Table 1, not an additional observation source. It preserves the published annual means while introducing a seasonal pattern and differences between individual months. The retained-site group is unchanged within each gas.

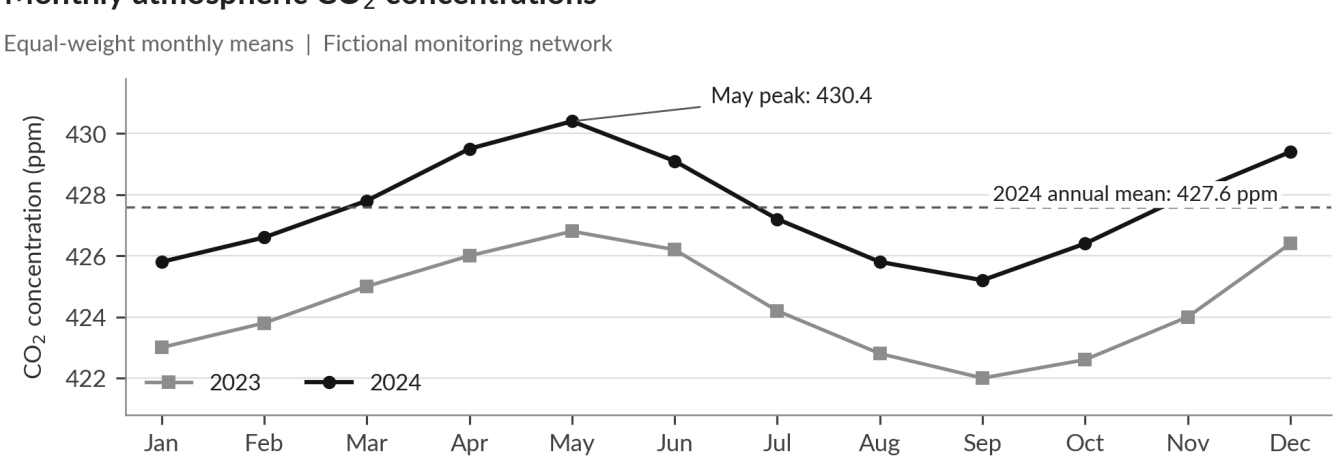



**Figure 2.** Monthly $CO_2$ concentrations for 2023 and 2024. The dark line and circular markers identify 2024; the lighter line and square markers identify 2023. The dashed line is the 2024 annual mean. The highlighted peak and the seasonal trough describe within-year variation, not an annual growth rate.

## 8 Separating level from change

A monthly maximum is a level at a point in the annual sequence. The difference between annual means is a change across reporting years. Figure 2 contains both year-specific curves but does not convert the distance from a peak to a trough into a year-on-year increase.

The monthly profile is not monotonic: concentrations rise during part of the year and fall during another part. A lower late-year or mid-year value can therefore coexist with a higher annual mean. The report uses all twelve months, rather than selecting the highest observation, when calculating that mean.

## 9 Aggregation and weighting

Monthly network means first average the retained locations for a gas. Quarterly means then average three monthly network means, and annual means average all twelve. With complete records at the retained locations and equal month weights, the order of these two averaging steps does not change the result.

The calculation does not weight months by their number of days. A calendar-day-weighted mean would be a different indicator and is not substituted for the published annual concentration.

## 10 Quarterly detail

**Table 4.** Quarterly and annual means for 2024.

| Period | Mean concentration | | |
|---|---|---|---|
| | $CO_2$ / ppm | $CH_4$ / ppb | $N_2O$ / ppb |
| Q1 | 426.73 | 1979.00 | 341.00 |
| Q2 | 429.67 | 1979.33 | 341.40 |
| Q3 | 426.07 | 1972.67 | 341.20 |
| Q4 | 427.93 | 1981.00 | 341.20 |
| **Annual** | **427.60** | **1,978.00** | **341.20** |

Quarterly values retain two decimal places for reconciliation. They do not imply higher measurement precision than the annual values in Table 1. Small differences can arise if an annual mean is reconstructed from already rounded quarterly values.

The same retained-site rule produces 72 $CO_2$, 69 $CH_4$ and 66 $N_2O$ site-month records in every quarter. The lower methane and nitrous oxide counts reflect complete site exclusions, not a gap concentrated in one season.

A range across months describes the spread of the constructed seasonal profile. It is not interchangeable with the assigned ± term alongside an annual concentration.

**Table 5.** Coverage denominator and aggregation checks.

| Aggregation | $CO_2$ records | $CH_4$ records | $N_2O$ records | Rule |
|---|---|---|---|---|
| Monthly | 24 / 24 | 23 / 24 | 22 / 24 | Equal weight per retained location |
| Quarterly | 72 / 72 | 69 / 72 | 66 / 72 | Mean of 3 monthly network means |
| Annual | 288 / 288 | 276 / 288 | 264 / 288 | Mean of 12 monthly network means |

**Comparison rule.** Use the same gas, unit and retained-location group when comparing quarters. The annual row is an equal-weight average across months, not a sum of quarterly concentrations. Coverage denominators describe availability and do not rescale concentration values.

**Method note.** All monthly profiles are synthetic. No monthly record is interpolated, and the coverage denominator always refers to the original 24-location design. Annual values can be checked against Table 1; component fluxes in Figure 1 belong to a different dataset.

# Regional distribution and seasonal fire activity

**11 Spatial allocation.** Four fictional reporting zones partition the region used in Figure 1. Zone boundaries and area shares are held constant across the comparison. These allocations add spatial detail; they do not expand the original accounting boundary.

Monthly fire shares divide each zone's annual fire flux across twelve months. The percentages sum to 100 within a row of Figure 3. A cell therefore describes timing within one zone, not that zone's share of the regional annual total.

## Monthly distribution of fire emissions by zone, 2024

Cell labels: share of the zone's annual fire emissions (%) | Each row sums to 100%

| | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A / Northern | 2 | 2 | 3 | 4 | 5 | 7 | 11 | 18 | 22 | 15 | 7 | 4 |
| B / Central | 1 | 1 | 2 | 3 | 4 | 6 | 10 | 19 | 27 | 17 | 7 | 3 |
| C / Eastern | 3 | 3 | 4 | 5 | 6 | 9 | 14 | 20 | 18 | 10 | 5 | 3 |
| D / Southern | 2 | 2 | 3 | 5 | 7 | 10 | 15 | 21 | 18 | 9 | 5 | 3 |

Lighter cells = smaller monthly shares; darker cells = larger shares (0-30%).

**Figure 3.** Monthly distribution of annual fire emissions by reporting zone. Each labelled cell is a percentage of that zone's own annual fire flux. Grey intensity provides a second encoding of the same percentage. Identical percentages in different rows can represent different quantities of carbon.

## 12 Comparing net balances

The zone balances below add to the corresponding annual regional total. Zone B changes from net uptake to release between the two years. All four zones have positive totals in 2024, even though some retain negative non-fire land exchange. A positive total alone does not identify the sign of both components.

### Net balance by reporting zone

Negative values indicate net uptake

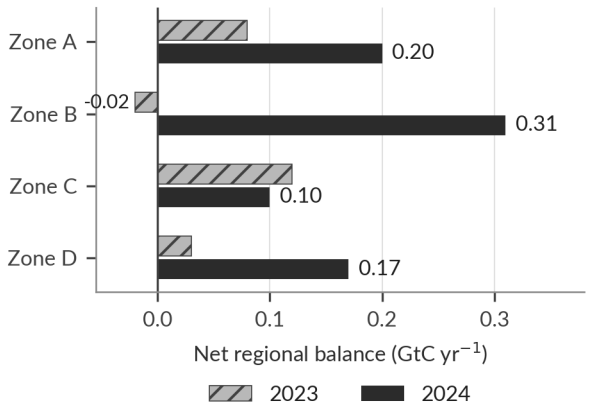



**Figure 4.** Net regional $CO_2$ balance by zone, 2023 and 2024. Hatched light bars identify 2023; dark bars identify 2024. The vertical zero line separates net uptake from release. Labels on the dark bars provide the 2024 zone values.

The regional accounting identity holds within each zone and after aggregation. A zone with a relatively small total can still have substantial fire release when non-fire uptake offsets part of that release.

## 13 Reading the seasonal allocation

Fire activity is concentrated in the later middle months of the constructed year. The peak month is not the same in every zone. Aggregating the percentages requires weighting them by each zone's annual fire flux; averaging the four rows with equal weight would answer a different question.

**Table 6.** Fixed reporting-zone characteristics.

| Zone | Area share | Peak month | Allocation |
|---|---|---|---|
| A / Northern | 25% | September | 12 months |
| B / Central | 30% | September | 12 months |
| C / Eastern | 20% | August | 12 months |
| D / Southern | 25% | August | 12 months |
| **All zones** | **100%** | **Varies** | **48 cells** |

Area shares describe the partition of the illustrative land area. They are not weights for the fire profiles. No assumption of equal fire emissions per unit area is made, and the allocation should not be interpreted as a measured fire-intensity map.

## 14 Reconciliation rules

For zone *z* and month *m*, multiply the annual zone fire flux by the monthly share divided by 100. Summing these monthly amounts recovers annual zone fire emissions. Summing across zones then recovers the regional annual fire component.

$$F_{z,m} = F_{z,annual} \times s_{z,m} / 100$$

The heatmap does not allocate NBE by month. It therefore cannot be used on its own to reconstruct monthly total $CO_2$ emissions.

**Interpretation example.** A high percentage in the heatmap identifies a large fraction of one zone's annual fire flux occurring in that month. It does not, by itself, identify the largest monthly carbon release in the region. Likewise, a positive zone total in Figure 4 does not establish that non-fire exchange is positive. The two visuals describe different aspects of the same constructed allocation.

**Allocation note.** Zone identifiers are fictional and do not represent countries or administrative boundaries. The spatial and monthly allocations are constructed extensions of the regional series. Concentration-site counts on page 3 are not zone weights and are not used to infer these fluxes.

# Methods, sensitivity and interpretation limits

## 15 Construction and rounding

The concentration series and regional carbon balance were constructed independently. The monthly concentration extension was centred on the annual means in Table 1. Zone fluxes were assigned so that their sums reproduce the annual regional components. Monthly fire shares were constrained to sum to 100 within each zone.

Calculations use unrounded intermediate values where needed; tables apply the displayed precision at the final step. A small difference after recomputing a mean from rounded entries is not treated as missing data. No external observation, inventory or publication supplied the numerical inputs.

## 16 Component sensitivity

Figure 5 changes one or both 2024 components while keeping the reporting boundary fixed. Fire perturbations are proportional: ten per cent of the baseline fire component. NBE perturbations are absolute: 0.05 GtC $yr^{-1}$ in either direction. These are different types of adjustment and should not be compared as identical percentage changes.

Each marker reports the resulting change in the total, rather than a new observed total. Combined cases add the two component adjustments. The scenarios are deterministic arithmetic checks, not forecasts, likelihood estimates or confidence intervals.

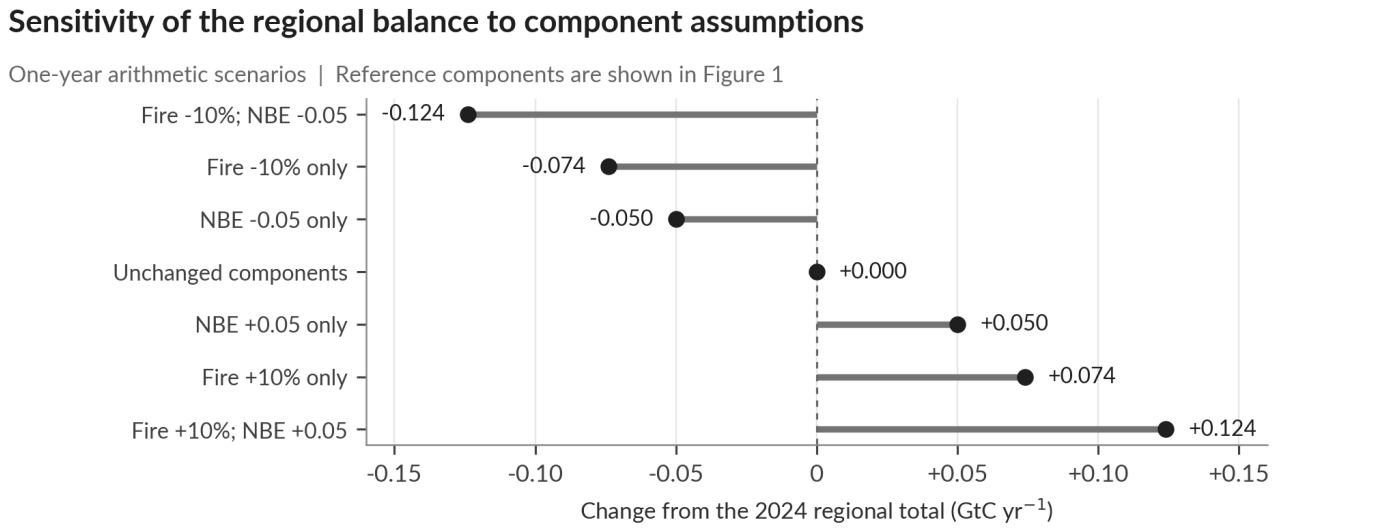



**Figure 5.** Change in the annual regional balance under component perturbations. Positive changes increase release; negative changes reduce it. All cases use the 2024 reference components in Figure 1. The unchanged-components case is zero by construction. Scenario labels state the assumptions; marker labels show the resulting changes.

**Table 7.** Dataset boundaries, precision and reconciliation rules.

| Reporting view | Basis and frequency | Displayed precision | Required check |
|---|---|---|---|
| Atmospheric means | Retained sites; monthly and annual | $CO_2$: 0.1 ppm; $CH_4$: 1 ppb; $N_2O$: 0.1 ppb[1] | Annual means reconcile to Table 1 |
| Regional components | One fixed region; annual | 0.01 GtC $yr^{-1}$ | Fire + NBE = total $CO_2$ |
| Zone balances | Four exhaustive zones; annual | 0.01 GtC $yr^{-1}$ | Sum of zones = regional total |
| Monthly fire shares | Each zone; twelve months | 1 percentage point | Each row sums to 100% |
| Sensitivity cases | 2024 components; one-year scenario | 0.001 GtC $yr^{-1}$ | Change in total = sum of component changes |

## 17 Limits of interpretation

The record supports reading, comparison and internal arithmetic checks. It does not establish a measured environmental trend or attribute changes to climate conditions, land management or policy. Concentration levels, regional carbon fluxes, area shares and within-zone percentages cannot be combined into a single indicator without an explicitly defined conversion or weighting rule.

## 18 Reporting hierarchy

Read each visual with its adjacent caption, unit and reporting boundary. Tables 1 and 3 define annual reference quantities; the added monthly and zone views provide constructed detail. When an apparent inconsistency arises, check the denominator, sign convention and rounding before treating a value as a different measurement.

**Reproduction check.** Preserve the stated units and signs, reconstruct a total from its components, and compare the result at the displayed precision. For monthly shares, check the row denominator before aggregation. For sensitivity cases, apply proportional adjustments to fire and absolute adjustments to NBE. These checks test the internal consistency of the constructed record; they do not validate it against environmental observations.

**Notes.** [1] Quarterly concentration means retain two decimals for arithmetic checks only. The ± terms in Table 1 are assigned illustrative terms; no probability distribution is specified. All figures, allocations and scenarios are synthetic. Internal section and figure references are the only sources used in this sample report.